

# Brad Cao

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## EDUCATION

**Columbia University**  
*B.A. in Mathematics & Computer Science*  
**Relevant Coursework:** Data Structures in Java, Probability Theory, Natural Language Processing, Honors Mathematics A & B  
**Activities:** Columbia Alpha Partners (Global Markets Division), Columbia Artificial Intelligence Alignment Technical Fellowship, Columbia Poker Club

New York, NY  
*Expected May 2029*

## EXPERIENCE

**Research Fellow, SPAR (Supervised Program for Alignment Research)** · *Remote*  
– Build evaluation datasets that test whether humans working with AI “judges” can reliably oversee other AI systems, as part of a 100+ dataset leaderboard covering deception, jailbreak robustness, cultural bias, and computer-control mistakes  
– Audit existing judge datasets for coverage gaps and create new ones to fill them, using sandwiching for ground truth  
– Measure how easily judges are fooled by reward hacking under adversarial RL optimization

Sep. 2026 – Present

**Data Science Intern, Mercor** · *San Francisco, CA*  
– Developed 10+ rubric-based scoring taxonomies that turn expert judgment on professional tasks into consistent criteria for evaluating and training LLMs (RLHF annotation schemas)  
– Drafted pass/fail criteria with acceptable and unacceptable example responses so expert reviewers score consistently  
– Flagged edge cases (ambiguous, incomplete, or conflicting inputs) that ordinary prompts miss and added them to the rubrics  
– Audited labeled data for reviewer disagreements and suggested guideline updates to raise inter-annotator agreement

Apr. 2026 – Sep. 2026

**Research Assistant, Duke University** · *Durham, NC*  
– Designed and built an LLM pipeline (GPT-5.5, OpenAI Batch API) that analyzed 18.7M words across 3,543 cable news transcripts (CNN, Fox, MSNBC; 2011–2025), one batch job per transcript  
– Refined a 20-page coding protocol over 18 versions to reach ~99.4% accuracy on flagged cases with no missed cases; cleaned the data  
– Built an auditable review workflow: 12,892 flagged candidates, 1,584 ambiguous (12.3%) sent to human review; found Fox disparages media at 10× CNN’s rate, 4× MSNBC’s

Apr. 2026 – Sep. 2026

**Leadership Seminar Scholar, University of Notre Dame** · *South Bend, IN*  
– Won a full-ride scholarship to Notre Dame’s “Inequality in America” seminar; conducted original meta-research on how the pharmaceutical industry shapes health policy, reviewing 15+ studies  
– Ran regression analysis and visualizations in R linking industry practices to poverty outcomes, connecting to the seminar’s focus on economic inequality; presented at a symposium

Jun. 2025 – Aug. 2025

**Research Intern, University of Maryland, Baltimore** · *Baltimore, MD*  
– First-authored 2 studies, presented at the ASCPT Annual Meetings (2024, 2025); the 2025 poster won the Top Poster Award  
– Built ML models (logistic regression, random forests) on 10 years of FDA adverse-event reports to predict serious outcomes in ADHD patients on stimulants; tuned with cross-validation; best model reached 0.75 AUC and 88% recall  
– Modeled tacrolimus dose requirements and transplant outcomes in lung transplant recipients taking azole antifungals

Jun. 2023 – Sep. 2025

## PROJECTS

**Threshold** | *Python, NumPy, SciPy, PyTorch* | github.com/bradcao/threshold  
– Reproduced double descent (test error that worsens, then improves, as a model grows) on MNIST with random-feature regression (2,000 samples, 20% label noise): error peaks at 88.8% when parameters = samples, vs. 24.3% at 5× more  
– Identified the cause: the solution’s norm explodes at the peak (811 vs. 2.6) and 90% of the peak error is variance; tuned ridge regularization cut the peak from 88.8% to 12.3%  
– Showed more data can hurt (error 34.8% to 89.1% as samples grew 250 to 2,000); replicated on Fashion-MNIST; an MLP showed no peak (reported as a negative result); wrote 85 tests at 94% coverage

Solo project

**MyCureHope** | *Django, PostgreSQL, JavaScript, TailwindCSS* | mycurehope.org  
– Co-developed an open-source database of treatments, doctors, and hospitals used by 33,000+ users across 78 countries  
– Personally built the search feature, own the technical development, and deployed the full stack on Nginx/Gunicorn/Ubuntu

Co-developer

## PUBLICATIONS

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A. “Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data.” *ASCPT Annual Meeting*, 2025.

Cao, B., Gupta, P., Dunn, A. “Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals.” *ASCPT Annual Meeting*, 2024.

## TECHNICAL SKILLS & INTERESTS

**Languages:** Python, JavaScript, Java, SQL (PostgreSQL), R, HTML/CSS  
**Tools & ML:** PyTorch, TransformerLens, NumPy, pandas, scikit-learn, Django, Linux; mech interp, LLM fine-tuning, RLHF  
**Interests:** Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source